Matthew T. Scoggins

mts2188@columbia.edu | mscoggs.github.io | mscoggs | 0000-0002-0748-9115

Education

Columbia University *New York, NY*
Ph.D. Candidate in Astronomy and Astrophysics *2021 – 2026 (expected)*
M.A. in Astronomy and Astrophysics (2023)
– **Advisors**: Zoltán Haiman and Greg Bryan
– **Thesis**: The Heavy Seed Pathway to Supermassive Black Holes

Western Washington University *Bellingham, WA*
B.S. in Physics & Mathematics *2016-2020*
B.A. in Philosophy
Minors in Computer Science and Astronomy

Appointments

Graduate Researcher *New York, NY*
Columbia University *2021-*

Teaching Assistant *New York, NY*
Columbia University *2021-2023*

Undergraduate Researcher *Bellingham, WA*
Western Washington University *2016-2021*

Lab Teaching Assistant *Bellingham, WA*
Western Washington University *2017-2020*

Honors & Awards

- **Explore Computing Time** – 400,000 ACCESS CU: 2023-2024
- **Edith and Robert Fehr Fellowship** – Columbia: 2022-2023
- **Magna Cum Laude** – WWU, BS & BA: 2020
- **Material Science Undergraduate Research Grant** – WWU: 2019
- **Oscar Edwin Olson Scholarship** – WWU: 2018-2020
- **Willard A. and Anne W. Brown Astronomy Scholar** – WWU: 2018
- **Summer Student Research Stipend** – WWU: 2018

Software

merger_tree_exploration *Github*
Code that analyzes Monte Carlo dark matter halo merger trees and searches for halos that may host direct-collapse black holes.

first_stars *Github*
Scripts that set up an ENZO simulation, run halo finding and merger trees, as well as find the critical mass for halo collapse.

lunar_impacts *Github*
Code used to set up a suite of simulations that run NASA's MEM3 code, as well as process and analyze their outputs.

star_lifting *Github*

A MESA wrapper which evolves the star with a time-dependent mass loss rate, keeping flux on a habitable planet constant.

qubit_simulation *Github*

Simulating the evolution of a superconducting chip with the goal of finding patterns in the optimal protocols (values of the controls over time which evolve an initial state into a target state in the shortest possible time) over a variety of initial and target combinations.

no_wave_qm *Github*

Simulating the evolution according to a Hamilton-Jacobi formulation of QM which replaces the wave with a configuration space density and equations of motion. Trajectory tracking using a 4-th order Runge Kutta technique.

Advising

Undergraduate Students

- Searching for Unusual Photometry in GAIA Data, 2023
  - **Andrea Dubbels** Columbia University Undergraduate.

Columbia STAR program

- Micrometeoroid Impact Rate Analysis for an Artemis-Era Lunar Base, 2024 – 2025.
  - **Nasiah Anderson,** CSS 12th Grade, Currently a Lafayette College Undergraduate
  - **Mark Driker,** CSS 11th Grade.
  - **Kokoro Onuma,** CSS 10th Grade.
- Determining the scatter in virial temperature for atomic cooling halo collapse, 2025 – 2026.
  - **Group will be assigned on Nov. 1st, 2025**

Lumiere

- Students took part in 2- to 12-month projects designed to expose them to research, typically within astronomy. Some projects have been (or will be) submitted to high school journals.

Teaching, Service & Outreach

– **Journal Referee:** ApJ, A&A *2023–*

– **Associate Director:** STAR Program *2023–*

– **Research Mentor:** STAR Program *2023 –*

– **Research Mentor:** Lumiere *2023–2025*

– **Mathematics TA, Physics TA, Physics Study Group Lead:** WWU *2017–2021*

Publications

Citations: 237 total. *h-index:* 6. 10 first-author papers, 4 co-author papers (4 in prep, links to drafts included when possible). NASA ADS link. Updated October 2025.

First Author Publications (* marks equal contributions)

10. **Scoggins, M. T.**, Haiman, Z., Bryan, G., & Kulkarni, M, 2025, Measuring the influence of merger histories on the formation of the first stars, (in prep, analysis complete)
9. **Scoggins, M. T.**, Haiman, Z., & Paucci, F, 2025, Heavy Seed Survivors in Dwarf Galaxies: A Case Study of Leo I, (in prep, expected on arXiv by November 2025. Draft)
8. **Scoggins, M. T.**, & Kipping, D., 2025, Lazarus Stars: Unique Signatures of Engineered Stars, (in prep, expected on arXiv by November 2025. Draft)
7. *Yahalomi, D., ***Scoggins, M. T.**, Anderson, N., Driker, M., et al., 2025, Meteoroid Impact Rate Analysis for an Artemis-Era Lunar Base, (in prep, expected on arXiv by November 2025. Draft)
6. **Scoggins, M. T.**, Ho, M., Villaescusa-Navarro, F., Jamieson, D., et al., 2025, Learning the Universe: 3 Gpc/h Tests of a Field Level N-Body Simulation Emulator, arXiv:2502.13242
5. **Scoggins, M. T.**, & Haiman, Z., 2024, Diagnosing the Massive-Seed Pathway to High-Redshift Black Holes: Statistics of the Evolving Black Hole to Host Galaxy Mass Ratio, MNRAS, **531**, 4584
4. **Scoggins, M. T.**, & Kipping, D., 2023, Lazarus Stars: Numerical Investigations of Stellar Evolution With Star-Lifting as a Life Extension Strategy, MNRAS, **523**, 3251
3. **Scoggins, M. T.**, Haiman, Z., & Wise, J., 2023, How Long Do High Redshift Massive Black Hole Seeds Remain Outliers in Black Hole Versus Host Galaxy Relations?, MNRAS, **519**, 2155
2. **Scoggins, M. T.**, & Rahmani, A., 2021, Topological and Geometric Patterns in Optimal Bang-Bang Protocols for Variational Quantum Algorithms: Application to the X X Z Model on the Square Lattice, Physical Review Research, **3**, 43165
1. **Scoggins, M. T.**, Davenport, J., & Covey, K., 2019, Using Flare Rates to Search for Stellar Activity Cycles, Research Notes of the American Astronomical Society, **3**, 137

Independent Significant Contribution

I contributed significant ideas, wrote/ran code, analyzed results, and/or wrote part of the manuscript.

4. Onoue, M., Ding, X., Silverman, J., Matsuoka, Y., et al. (including **Scoggins, M. T.**), 2025, A Post-Starburst Pathway for the Formation of Massive Galaxies and Black Holes at z>6, Nature Astronomy
3. Li, J., Silverman, J., Shen, Y., Volonteri, M., et al. (including **Scoggins, M. T.**, 2025, Tip of the Iceberg: Overmassive Black Holes at 4<z<7 Found by JWST Are Not Inconsistent With the Local Relations, Apj, **981**, 19
2. Roser, P., & **Scoggins, M. T.**, 2023, Non-Quantum Behaviors of Configuration-Space Density Formulations of Quantum Mechanics, arXiv:2303.04959
1. Olney, R., Kounkel, M., Schillinger, C., **Scoggins, M. T.**, et al., 2020, APOGEE Net: Improving the Derived Spectral Parameters for Young Stars Through Deep Learning, AJ, **159**, 182